SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Performance Testing

Date	2 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing
Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter 5.6.3
Type of Testing	Load Testing
Target Module / API	Home Page (HTTP GET Request) of Credit Card Approval Prediction Web Application
Test Environment	Local System (Windows 10, Flask Application, Python 3.13)
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Home Page	20	10	Page should load successfully without errors
2	Multiple Concurrent Requests	20	10	All requests should be processed successfully
3	Response Time Measurement	20	10	Average response time should remain low
4	Throughput Evaluation	20	10	Application should handle multiple requests efficiently

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	5 ms	Pass	Very fast response time
2	Response Time (Max)	< 5 seconds	40 ms	Pass	Maximum response time well within the limit
3	Throughput (Req/sec)	> 10 Req/sec	10.6 Req/sec	Pass	Stable request handling
4	Error Rate	< 1%	0.00%	Pass	No request failures
5	CPU Utilization	< 80%	Below 80% (Normal)	Pass	System remained stable during testing
6	Memory Utilization	< 80%	Below 80% (Normal)	Pass	No excessive memory usage observed

Step 4: Observations & Analysis

Key Findings

The Credit Card Approval Prediction System was evaluated using Apache JMeter 5.6.3 under load testing conditions. The application successfully processed 100 HTTP requests generated by 20 virtual users. The average response time was 5 milliseconds, while the maximum response time recorded was 40 milliseconds. The system achieved a throughput of 10.6 requests per second with an error rate of 0.00%, indicating reliable and efficient performance.

Bottlenecks Identified

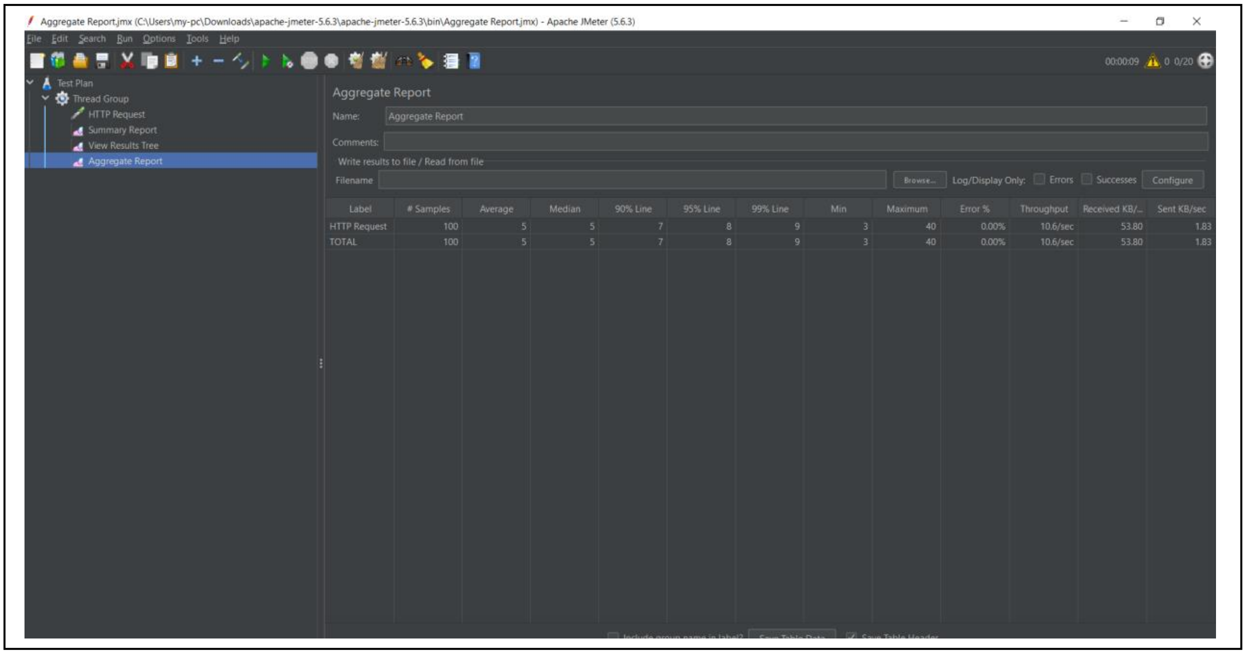
No significant performance bottlenecks were observed during testing. The application maintained consistent response times throughout the test, and all requests were completed successfully without failures. The measured response times remained well within the expected limits.

Optimization Steps Taken

The application performance was improved through effective data preprocessing, feature engineering, optimized machine learning models, and a lightweight Flask web application. These optimizations enabled fast prediction responses and stable application performance during load testing.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):


Aggregate Report.jmx (C:\Users\my-pc\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\Aggregate Report.jmx) - Apache JMeter (5.6.3)
File Edit Search Run Options Tools Help
Test Plan
Thread Group
HTTP Request
Summary Report
View Results Tree
Aggregate Report
Aggregate Report
Name: Aggregate Report
Comments:
Write results to file / Read from file
Filename
Browse...
Log/Display Only:
Errors
Successes
Configure
Label | # Samples | Average | Median | 90% Line | 95% Line | 99% Line | Min | Maximum | Error % | Throughput | Received KB/... | Sent KB/sec
HTTP Request | 100 | 5 | 5 | 7 | 8 | 9 | 3 | 40 | 0.00% | 10.6/sec | 53.80 | 1.83
TOTAL | 100 | 5 | 5 | 7 | 8 | 9 | 3 | 40 | 0.00% | 10.6/sec | 53.80 | 1.83


Credit Card Approval Prediction
Applicant ID
Gender
Male
Owned Car
Yes
Owned Realty
Yes
Total Children
Total Income
Income Type
Working
Education
Higher education
Family Status
Married
Housing Type
House / apartment
Mobile Phone
Yes
Work Phone
Yes

Phone
Yes

Email
Yes

Job Title
Managers

Total Family Members

Applicant Age

Years of Working

Total Bad Debt

Total Good Debt

Predict Credit Card Approval